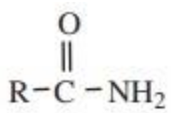
Acid Amides

Acid amides are derivatives of carboxylic acid in which the – OH part of carboxyl group has been replaced by – NH_2 group.

The general formula of amides are given as follows.

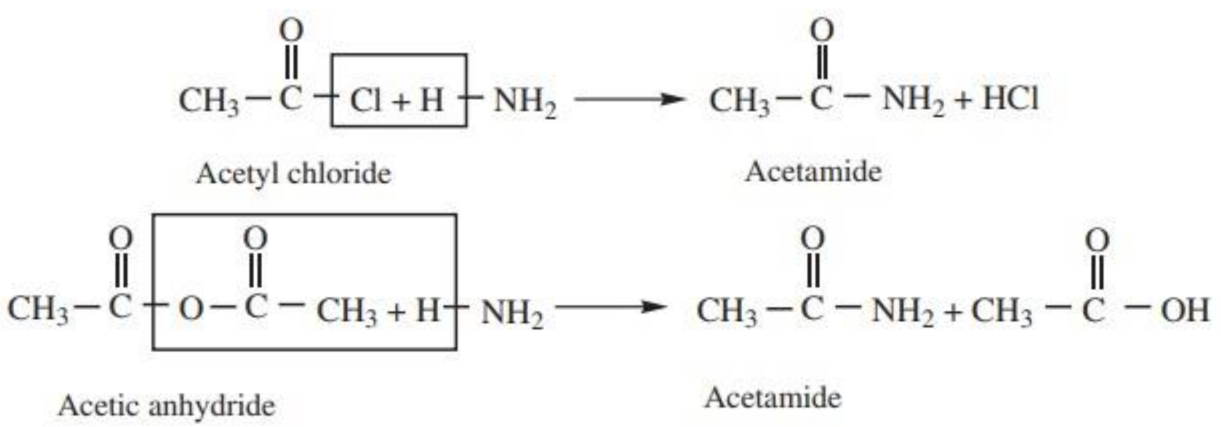


Now, we shall focus our attention mainly on the study of chemistry of acetamide.

Methods of Preparation

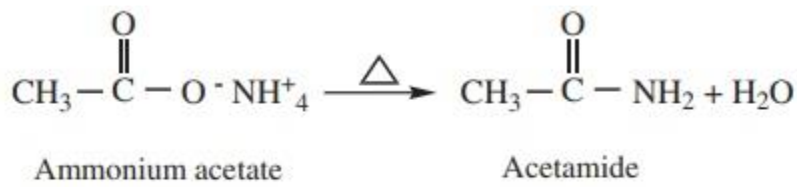
1. Ammonolysis of acid derivatives

Acid amides are prepared by the action of ammonia with acid chlorides or acid anhydrides.



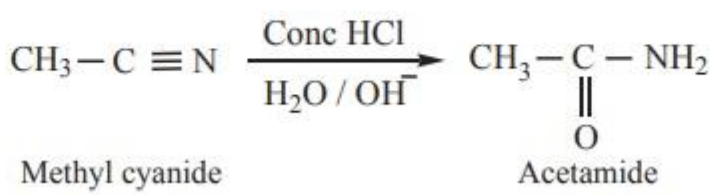
2) Heating ammonium carboxylates

Ammonium salts of carboxylic acids (ammonium carboxylates) on heating, lose a molecule of water to form amides.



3) Partial hydrolysis of alkyl cyanides (Nitriles)

Partial hydrolysis of alkyl cyanides with cold con HCl gives amides



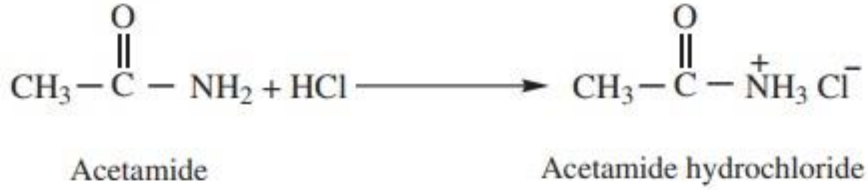
Chemical Properties

1. Amphoteric character

Amides behave both as weak acid as well as weak base and thus show amphoteric character.

This can be proved by the following reactions.

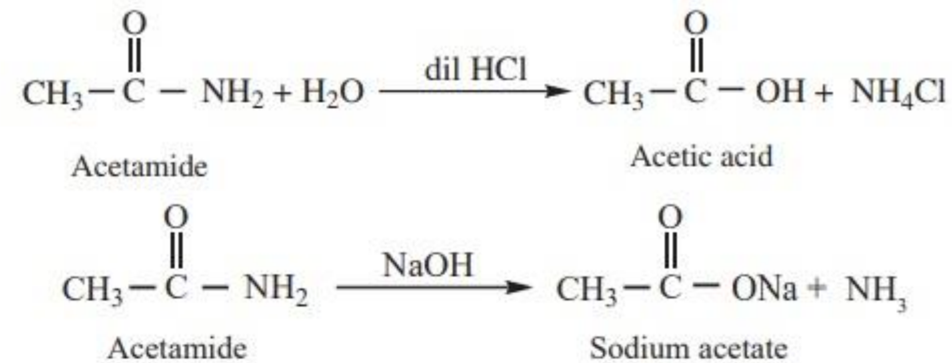
Acetamide (as base) reacts with hydrochloric acid to form salt



Acetamide (as acid) reacts with sodium to form sodium salt and hydrogen gas is liberated.

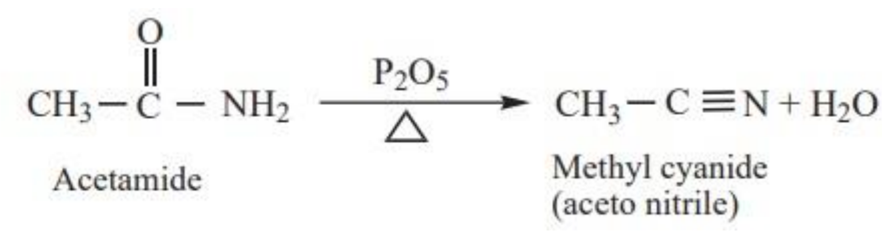
2) Hydrolysis

Amides can be hydrolysed in acid or in alkaline solution on prolonged heating



3) Dehydration

Amides on heating with strong dehydrating agents like P_2O_5 get dehydrated to form cyanides.



4) Hoff mann’s degradation

Amides reacts with bromine in the presence of caustic alkali to form a primary amine carrying one carbon less than the parent amide.

$$\underset{\text{Acetamide}}{CH_3-\overset{\overset{\displaystyle O}{\|}}{C}-NH_2} + Br_2 + 4\,KOH \xrightarrow{\triangle} \underset{\text{Methyl amine}}{CH_3NH_2} + K_2CO_3 + 2KBr + 2H_2O$$

5) Reduction

Amides on reduction with $LiAlH_4$or Sodium and ethyl alcohol to form corresponding amines.

$$\underset{\text{Acetamide}}{CH_3-\overset{\overset{\displaystyle O}{\|}}{C}-NH_2} + 4\,(H) \xrightarrow{LiAlH_4} \underset{\text{Ethyl amine}}{CH_3-CH_2-NH_2} + H_2O$$

Azhakarajpootkhokhar@gmail.com